# Review

A linear system of $m$ equations and $n$ variables can be written as

$$A\mathbf{x} = \mathbf{b}$$

where $A$ is $m \times n$, $\mathbf{x}$ is $n \times 1$, and $\mathbf{b}$ is $m \times 1$.

## Example

$$\begin{cases} y_1 & -2y_2 & & +y_4 = -2 \\ & -3y_2 & +5y_3 & -3y_4 = -6 \\ -y_1 & -7y_2 & & -y_4 = 3 \end{cases} \rightsquigarrow \begin{bmatrix} 1 & -2 & 0 & 1 \\ 0 & -3 & 5 & -3 \\ -1 & -7 & 0 & -1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} -2 \\ -6 \\ 3 \end{bmatrix}$$

# Vector parametric form (review)

The solution of a consistent linear system can be expressed in vector parametric form as

$$\mathbf{x} = \mathbf{v}_0 + x_{i_1}\mathbf{v}_1 + x_{i_2}\mathbf{v}_2 + \cdots + x_{i_r}\mathbf{v}_r$$

where $x_{i_1}, \ldots, x_{i_r}$ are the free variables.

## Example

Solve the linear system $\begin{cases} -x_1 + 2x_2 + x_3 = 1 \\ 2x_1 - 4x_2 = -4 \\ -x_1 + 2x_2 + 3x_3 = -1 \end{cases}$

The augmented matrix and its R.E.F. are :

$$\left[\begin{array}{ccc|c} -1 & 2 & 1 & 1 \\ 2 & -4 & 0 & -4 \\ -1 & 2 & 3 & -1 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & -2 & 0 & -2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right] \rightsquigarrow \begin{cases} x_1 - 2x_2 = -2 \\ x_3 = -1 \\ 0 = 0 \end{cases}$$

The vector parametric form of the general solution is :

$$\begin{cases} x_1 = 2x_2 - 2 \\ x_2 : \text{free} \\ x_3 = -1 \end{cases} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2x_2 - 2 \\ x_2 \\ -1 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ -1 \end{bmatrix} + \begin{bmatrix} 2x_2 \\ x_2 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

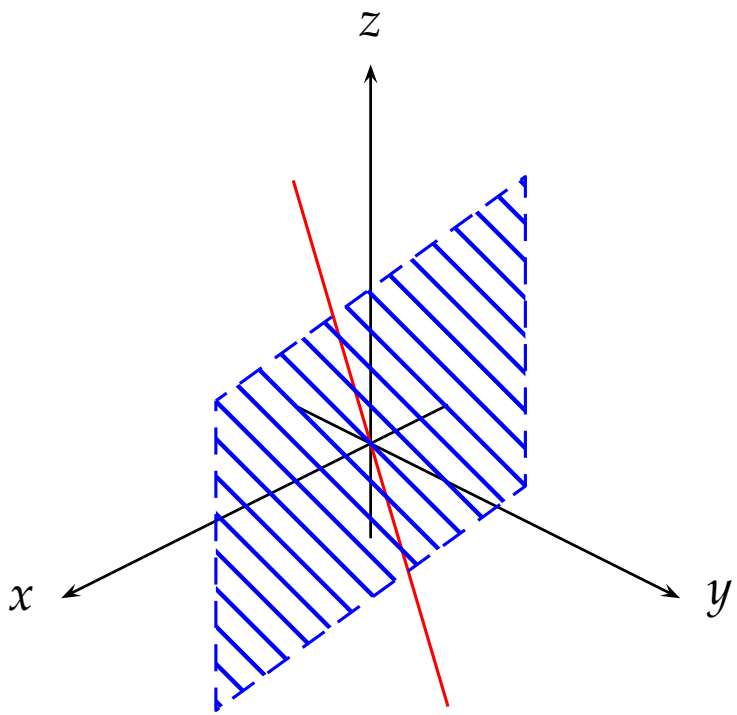
# Review of homogeneous linear systems

- A linear system $A\mathbf{x} = \mathbf{b}$ is called homogeneous if $\mathbf{b} = \mathbf{0}$.
- Equivalently, a homogeneous system can be written as the matrix equation $A\mathbf{x} = \mathbf{0}$.
- A homogeneous system always has a trivial solution $\mathbf{x} = \mathbf{0}$, i.e., $x_1 = 0, x_2 = 0, \ldots$
- Threfore, a homogeneous system is always consistent.
- A solution $\mathbf{x} \neq 0$ of a homogeneous system is called a nontrivial solution.
- A homogeneous system has ...
  - a unique solution if all variables are basic.
  - infinitely many solutions if there exists at least one free variable.

• A homogeneous linear system has nontrivial solutions if and only if it has free variables.

# Motivation

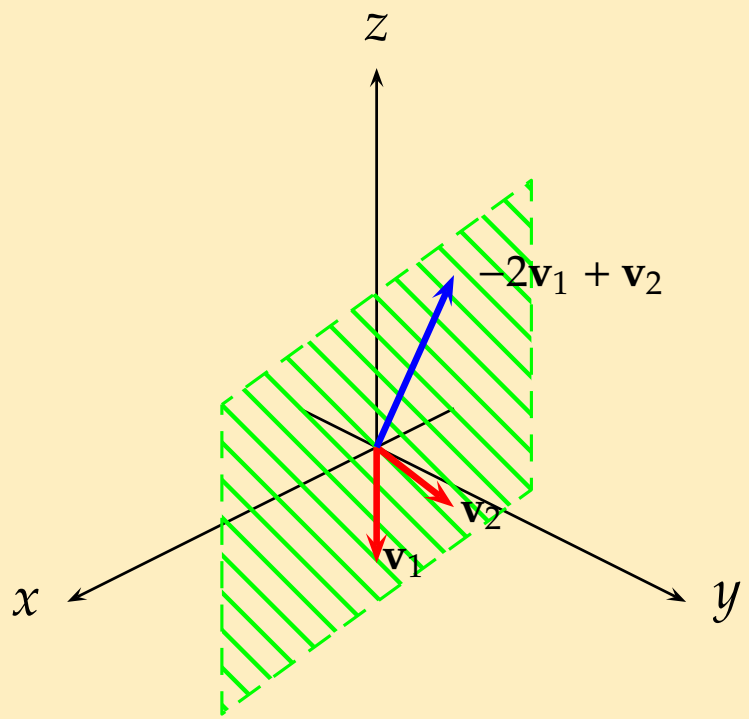
Question. What is the difference between a line and a plane?




**Answer 1.** A line is one-dimensional, whereas a plane is two-dimensional.

**Answer 2.** On a line there is only one independent direction. On a plane there are two independent directions.

# Geometric meaning...




- If we fix two non-parallel vectors $\mathbf{v}_1, \mathbf{v}_2$ in a plane, then every vector in the plane is a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$.

- In other words, every vector in the plane is in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$.

## Linear dependence/independence

Suppose $\mathbf{v}_1, ..., \mathbf{v}_p$ are vectors in $\mathbb{R}^n$.

The vector equation

$$x_1\mathbf{v}_1 + \cdots + x_p\mathbf{v}_p = \mathbf{0} \qquad (\star)$$

always has a trivial solution $x_1 = \cdots = x_p = 0$.

- The vectors $\mathbf{v}_1, ..., \mathbf{v}_p$ are said to be linearly dependent if the equation $(\star)$ has a nontrivial solution. (That is, a solution with *at least* one $x_i \neq 0$.)
- The vectors $\mathbf{v}_1, ..., \mathbf{v}_p$ are said to be linearly independent if the equation $(\star)$ has no nontrivial solutions.

**Example.** The vectors $\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$, and $\begin{bmatrix} 1 \\ -5 \\ 4 \end{bmatrix}$ are linearly dependent because

$$\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} + 2\begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} + (-1)\begin{bmatrix} 1 \\ -5 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

## Linear dependence/independence

Suppose $\mathbf{v}_1, ..., \mathbf{v}_p$ are vectors in $\mathbb{R}^n$.

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- The vectors $\mathbf{v}_1, ..., \mathbf{v}_p$ are said to be linearly independent if the equation $(\star)$ has no nontrivial solutions.

**Example.** The vectors $\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, and $\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$ are linearly dependent because:

$$\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + (0)\begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + (-\frac{1}{2})\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

## Linear dependence/independence

Suppose $\mathbf{v}_1, \ldots, \mathbf{v}_p$ are vectors in $\mathbb{R}^n$.

The vector equation

$$x_1\mathbf{v}_1 + \cdots + x_p\mathbf{v}_p = \mathbf{0} \qquad (\star)$$

always has a trivial solution $x_1 = \cdots = x_p = 0$.

- The vectors $\mathbf{v}_1, \ldots, \mathbf{v}_p$ are said to be linearly dependent if the equation $(\star)$ has a nontrivial solution. (That is, a solution with *at least* one $x_i$ nonzero.)
- The vectors $\mathbf{v}_1, \ldots, \mathbf{v}_p$ are said to be linearly independent if the equation $(\star)$ has no nontrivial solutions.

**Example.** The vectors $\begin{bmatrix}1\\0\\0\\1\end{bmatrix}$, $\begin{bmatrix}0\\1\\0\\-2\end{bmatrix}$, and $\begin{bmatrix}0\\0\\1\\-3\end{bmatrix}$ are linearly independent, because by looking at the entries we have:

$$x_1\begin{bmatrix}1\\0\\0\\1\end{bmatrix} + x_2\begin{bmatrix}0\\1\\0\\-2\end{bmatrix} + x_3\begin{bmatrix}0\\0\\1\\-3\end{bmatrix} = \begin{bmatrix}x_1\\x_2\\x_3\\x_1 - 2x_2 - 3x_3\end{bmatrix} = \begin{bmatrix}0\\0\\0\\0\end{bmatrix} \Rightarrow x_1 = x_2 = x_3 = 0$$

Checking linear dependence/independence amounts to checking if a vector equation has a nontrivial solution or not.

**Example**

Are the vectors $\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ -3 \end{bmatrix}$ linearly dependent or independent?

- The question is equivalent to determining if the vector equation $x_1 \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 1 \\ -3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ has a nontrivial solution.
- Next we row reduce the augmented matrix :

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 1 & -1 & 1 & 0 \\ -1 & 2 & -3 & 0 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \text{R.E.F.}$$

- $x_3$ is free, therefore the system has nontrivial solutions. Therefore the above vectors are linearly dependent.

Checking linear dependence/independence amounts to checking if a vector equation has a nontrivial solution or not.

**Example**

Are the vectors $\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ -3 \end{bmatrix}$ linearly dependent or independent?

- The question is equivalent to determining if the vector equation $x_1 \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 1 \\ -3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ has a nontrivial solution.
- General solution of the vector equation:

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 1 & -1 & 1 & 0 \\ -1 & 2 & -3 & 0 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \rightsquigarrow \begin{cases} x_1 = x_3 \\ x_2 = 2x_3 \\ x_3 : \text{free} \end{cases}$$

- **An explicit dependence relation:** Set $x_3 = 1$. Then $x_1 = 1$ and $x_2 = 2$ and we obtain the relation

$$\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \\ -3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Checking linear dependence/independence amounts to checking if a vector equation has a nontrivial solution or not.

**Example**

Are the vectors $\begin{bmatrix}-1\\0\\1\end{bmatrix}, \begin{bmatrix}1\\1\\2\end{bmatrix}, \begin{bmatrix}1\\2\\5\end{bmatrix}, \begin{bmatrix}0\\1\\3\end{bmatrix}$ linearly dependent or independent?

- The question is equivalent to determining if the vector equation $x_1\begin{bmatrix}-1\\0\\1\end{bmatrix}, +x_2\begin{bmatrix}1\\1\\2\end{bmatrix} + x_3\begin{bmatrix}1\\2\\5\end{bmatrix} + x_4\begin{bmatrix}0\\1\\3\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$ has a nontrivial solution.
- General solution of the vector equation:

$$\left[\begin{array}{cccc|c} -1 & 1 & 1 & 0 & 0\\ 0 & 1 & 2 & 1 & 0\\ 1 & 2 & 5 & 3 & 0 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{cccc|c} 1 & 0 & 1 & 1 & 0\\ 0 & 1 & 2 & 1 & 0\\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \rightsquigarrow \begin{cases} x_1 = -x_3 - x_4\\ x_2 = -2x_3 - x_4\\ x_3 : \text{free}\\ x_4 : \text{free} \end{cases}$$

  Therefore the vectors are **linearly dependent**.
- **An explicit dependence relation:** Set $x_3 = 1, x_4 = -1$. Then we obtain $x_1 = 0$, $x_2 = -1$ and therefore we have the relation

$$0\begin{bmatrix}-1\\0\\1\end{bmatrix} + (-1)\begin{bmatrix}1\\1\\2\end{bmatrix} + \begin{bmatrix}1\\2\\5\end{bmatrix} + (-1)\begin{bmatrix}0\\1\\3\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}.$$

## Example

Are the columns of the following matrix linearly dependent or independent?

$$A = \begin{bmatrix} 3 & -1 & 1 \\ 3 & 1 & 0 \\ 6 & -2 & -5 \end{bmatrix}$$

**Solution.**

- We write the augmented matrix of the corresponding homogeneous system:

$$x_1 \begin{bmatrix} 3 \\ 3 \\ 6 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 0 \\ -5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \left[\begin{array}{ccc|c} 3 & -1 & 1 & 0 \\ 3 & 1 & 0 & 0 \\ 6 & -2 & -5 & 0 \end{array}\right]$$

- We row reduce this matrix to obtain an E.F.

$$\left[\begin{array}{ccc|c} 3 & -1 & 1 & 0 \\ 3 & 1 & 0 & 0 \\ 6 & -2 & -5 & 0 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{ccc|c} 3 & -1 & 1 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 3 & 0 \end{array}\right] \quad \text{E.F.}$$

- Every column is a pivot column, so there are no free variables, i.e., the system has no nontrivial solution.
- It follows that the columns of the above matrix are linearly independent.

## Example

Are the columns of $A = \begin{bmatrix} 1 & 4 & -2 & -1 \\ 0 & -1 & -1 & -2 \\ -3 & -3 & 1 & -7 \\ 1 & 0 & 1 & 5 \end{bmatrix}$ linearly dependent or linearly independent?

**Solution**

$$\left[\begin{array}{cccc|c} 1 & 4 & -2 & -1 & 0 \\ 0 & -1 & -1 & -2 & 0 \\ -3 & -3 & 1 & -7 & 0 \\ 1 & 0 & 1 & 5 & 0 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 3 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \text{ R.E.F.}$$

There are free variables, therefore the linear system has nontrivial solutions, therefore the columns of $A$ are **linearly dependent**.

## Summary of the Algorithm

To determine if the vectors $\mathbf{v}_1, \ldots, \mathbf{v}_p$ in $\mathbb{R}^n$ are linearly dependent/independent, we first reduce the matrix

$$[\mathbf{v}_1 \ \cdots \ \mathbf{v}_p]$$

to an echelon form.

- If every column in E.F. contains a pivot position, then the vectors are linearly independent.
- If at least one of the columns in E.F. is a non-pivot column, then the vectors are linearly dependent.

Suppose you are given a vector $\mathbf{v}$ in $\mathbb{R}^n$.

Question. Is the set $\{\mathbf{v}\}$ linearly dependent or independent?

Answer. If $\mathbf{v} \neq \mathbf{0}$ then $\{\mathbf{v}\}$ is linearly independent.(Because if $x\mathbf{v} = \mathbf{0}$ then $x = 0$.)

However, the set $\{\mathbf{0}\}$ is linearly dependent, because for example $1.\mathbf{0} = \mathbf{0}$.

Suppose $\mathbf{v}_1, \mathbf{v}_2$ are nonzero vectors in $\mathbb{R}^n$.

Question. When is the set $\{\mathbf{v}_1, \mathbf{v}_2\}$ linearly dependent?

Answer. Exactly when $\mathbf{v}_1$ and $\mathbf{v}_2$ are parallel.

In fact suppose $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 = \mathbf{0}$ where $x_1 \neq 0$ or $x_2 \neq 0$.

If, say, $x_1 \neq 0$, then we have

$$x_1\mathbf{v}_1 + x_2\mathbf{v}_2 = \mathbf{0} \Rightarrow \mathbf{v}_1 = -\frac{x_2}{x_1}\mathbf{v}_2$$

which means $\mathbf{v}_1$ and $\mathbf{v}_2$ are parallel.

(The case $x_2 \neq 0$ is similar.)

Question. Suppose $\mathbf{v}$ is a nonzero vector in $\mathbb{R}^n$, and $\mathbf{0}$ denotes the zero vector in $\mathbb{R}^n$. Is the set $\{\mathbf{0}, \mathbf{v}\}$ linearly dependent or linearly independent?

Answer. Linearly dependent because for example:

$$1 \cdot \mathbf{0} + 0 \cdot \mathbf{v} = \mathbf{0}.$$

In other words the equation $x_1\mathbf{0} + x_2\mathbf{v} = \mathbf{0}$ has a nontrivial solution $x_1 = 1, x_2 = 0$.

Suppose $\mathbf{v}_1, ..., \mathbf{v}_p$ are vectors in $\mathbb{R}^n$. If at least one of the $\mathbf{v}_i$'s is zero, then $\mathbf{v}_1, ..., \mathbf{v}_p$ are linearly dependent.

**Example.** The vectors $\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, and $\begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$ are linearly dependent since

$$0\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + 2\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + 0\begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

## Characterization of linearly dependent sets

Suppose $\mathbf{v}_1, ..., \mathbf{v}_p$ are vectors in $\mathbb{R}^n$. Then :

- If $\mathbf{v}_1, ..., \mathbf{v}_p$ are linearly dependent then at least one of the $\mathbf{v}_i$'s is a linear combination of the other vectors.
- Conversely, if one of the $\mathbf{v}_i$'s is a linear combination of the other vectors, then $\mathbf{v}_1, ..., \mathbf{v}_p$ are linearly dependent.

**Example.** If we have

$$-3\mathbf{v}_1 + 2\mathbf{v}_2 - \mathbf{v}_3 = \mathbf{0}$$

then we can write

$$\mathbf{v}_1 = \frac{2}{3}\mathbf{v}_2 - \frac{1}{3}\mathbf{v}_3$$

or

$$\mathbf{v}_2 = \frac{3}{2}\mathbf{v}_1 + \frac{1}{2}\mathbf{v}_3$$

or

$$\mathbf{v}_3 = -3\mathbf{v}_1 + 2\mathbf{v}_2$$

**Theorem**

Suppose $\mathbf{v}_1, ..., \mathbf{v}_p$ are vectors in $\mathbb{R}^n$. If $p > n$, then $\mathbf{v}_1, ..., \mathbf{v}_p$ are linearly dependent.

Reason : Think of the linear system with augmented matrix

$$[\ \mathbf{v}_1\ \ \mathbf{v}_2\ \ \cdots \mathbf{v}_p \mid \mathbf{0}\ ]$$

- The matrix $A = [\ \mathbf{v}_1\ \ \mathbf{v}_2\ \ \cdots \mathbf{v}_p\ ]$ has at most $n$ pivot positions (because it has $n$ rows).
- The matrix $A = [\ \mathbf{v}_1\ \ \mathbf{v}_2\ \ \cdots \mathbf{v}_p\ ]$ has $p$ columns and $p > n$.
- Therefore there exists at least one non-pivot column, i.e., one free variable.
- Hence the system has nontrivial solutions.

**Theorem**

Suppose $\mathbf{v}_1, ..., \mathbf{v}_p$ are vectors in $\mathbb{R}^n$. If $p > n$, then $\mathbf{v}_1, ..., \mathbf{v}_p$ are linearly dependent.

**Example.** Are the vectors

$$\begin{bmatrix} 1 \\ -2 \\ 11 \\ -2 \end{bmatrix}, \begin{bmatrix} 3 \\ 5 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -3 \\ 3 \\ 5 \\ 7 \end{bmatrix}, \begin{bmatrix} -4 \\ 3 \\ 1 \\ 4 \end{bmatrix} \text{ and } \begin{bmatrix} 0 \\ 3 \\ -2 \\ 1 \end{bmatrix}$$

linearly dependent or independent?

**Answer.** They are linearly dependent because we have 5 vectors in $\mathbb{R}^4$ and $5 > 4$ !

**The punchline...**

If we choose more than $n$ vectors in $\mathbb{R}^n$, then they will be linearly dependent.

**Warning.** This does not mean that if we choose $n$ or fewer vectors in $\mathbb{R}^n$ then they will be linearly independent!

## Example

**Question.** We are given 13 vectors in $\mathbb{R}^5$. Can they be linearly independent?

**Answer.** No, they can't, because $13 > 5$.

**Question.** If we are given 3 vectors in $\mathbb{R}^5$, are they always linearly independent?

**Answer.** No, for example, if we are given

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

then

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Therefore the above vectors are linearly dependent.